

Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)			bright yellow / canary yellow	1			1		
	(b)			$\text{Pb}^{2+} + 2\text{I}^- \rightarrow \text{PbI}_2$	1			1		
2				$[\text{CuCl}_4]^{2-}$	1			1		
3	(a)			rate = $k[\text{N}_2\text{O}_5]$	1			1		
	(b)			accept any balanced equation that has one N_2O_5 as reactant e.g. $\text{N}_2\text{O}_5 \rightarrow \text{NO}_2 + \text{NO} + \text{O}_2$		1		1		
4				potential difference/EMF measured when a half-cell is connected to the standard hydrogen electrode (1) award (1) for any two of the standard conditions 298 K temperature 1 atm pressure 1 mol dm ⁻³ concentration	2			2		
5				award (1) for either of following phosphorus can expand octet but nitrogen cannot phosphorus has available d-orbitals (so can have more than 8 electrons in outer shell in molecules) but nitrogen does not answer must refer to both elements	1			1		

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					AO1	AO2	AO3	Total	Maths	Prac
6				$K_w = [\text{H}^+][\text{OH}^-]$	1			1		1
7				particles have greater freedom in liquid mercury compared to solid gold (so they have less order in liquid and higher entropy)	1			1		
				Section A total	9	1	0	10	0	1